

FINAL YEAR PROJECT PROPOSAL



RecruitSense: AI-Assisted Recruitment Decision Support System

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1 Introduction

1.1 Background of Recruitment Systems

Recruitment has become increasingly complex due to the rapid growth of online job platforms such as LinkedIn [5], Glassdoor [6], and Indeed [7], along with the large volume of applications received for a single job posting. In the information technology sector, recruiters often receive hundreds or even thousands of resumes, making manual screening time-consuming, inconsistent, and prone to human bias.

As the number of applicants increases, it becomes difficult to ensure fair, objective, and efficient evaluation of every candidate. Traditional manual shortlisting methods may overlook qualified candidates or result in inconsistent decision-making due to subjective judgment.

1.2 Limitations of Existing Recruitment Platforms

Existing recruitment platforms primarily focus on job posting, professional networking, and application collection [5, 6, 7]. While such platforms provide basic filtering and keyword-based search mechanisms, they do not offer structured decision-support mechanisms to help recruiters objectively compare candidates based on skills and job requirements.

Similarly, traditional Applicant Tracking Systems (ATS) rely mainly on keyword-based filtering techniques [8], which may fail to capture contextual meaning within resumes. As a result, qualified candidates may be overlooked due to rigid matching rules.

Job seekers, especially students and fresh graduates, receive little to no constructive feedback on their skill readiness or job suitability after submitting applications. Most platforms do not provide clear insight into missing skills or areas requiring improvement, limiting opportunities for structured self-development.

1.3 Proposed Solution Overview

This project proposes an AI-assisted recruitment decision support system that uses Natural Language Processing (NLP) techniques [1, 3] to analyze uploaded resumes. The system extracts skills, tools, and relevant experience from resumes and compares them with company-defined job requirements using text vectorization and similarity scoring techniques.

Specifically, TF-IDF vectorization and cosine similarity [2] are applied to compute alignment scores between candidate profiles and job descriptions. These techniques are widely used in information retrieval and text analysis for measuring document similarity.

The system identifies matching strengths and missing skills, providing structured ranking for recruiters and actionable feedback for job seekers. The objective of the proposed system is not to replace human decision-making, but to support recruiters with data-driven insights while simultaneously helping job seekers understand their strengths, weaknesses, and learning needs.

2 Objective

The objective of this project is to design and develop an AI-assisted recruitment decision support system that analyzes resumes using Natural Language Processing (NLP) techniques [1] and compares extracted candidate skills with company-defined job requirements through similarity-based evaluation [2].

The system also incorporates a quiz-based non-technical skill assessment module to support holistic candidate evaluation. It aims to assist recruiters in efficient shortlisting and ranking while providing job seekers with transparent feedback on job suitability and identified skill gaps.

3 Problem Description

3.1 Challenges Faced by Recruiters

Recruiters often receive a large volume of applications for a single job posting through on-line platforms such as LinkedIn [5] and Indeed [7], making manual resume screening time-consuming and inefficient. Evaluating candidates consistently becomes difficult due to subjective judgment and variations in resume formats.

Additionally, traditional Applicant Tracking Systems (ATS) rely heavily on keyword-based filtering mechanisms [8]. Comparing candidates objectively based on skills and experience therefore requires significant effort, often leading to delayed shortlisting decisions.

3.2 Challenges Faced by Job Seekers

Job seekers, particularly students and fresh graduates, lack clear insight into how well their skills align with specific job roles. Existing platforms such as Glassdoor [6] and LinkedIn [5] rarely provide structured or transparent feedback explaining why a candidate was not shortlisted or what skills are missing.

Moreover, candidates are often not evaluated on non-technical competencies in a standardized manner, leaving them uncertain about how to improve their overall job readiness.

3.3 Need for a Structured Decision Support System

Current systems rely heavily on basic keyword filtering [8], which may not effectively measure overall skill alignment and job suitability. There is a need for a structured recruitment decision support system that systematically analyzes resume content using NLP techniques [1], incorporates non-technical skill assessment, assists recruiters in ranking candidates through similarity-based evaluation [2], and provides meaningful, skill-based feedback without fully automating hiring decisions.

4 Methodology

4.1 Data Collection

A real-world resume dataset in PDF format covering multiple professional domains (including IT, Engineering, and Business) is used to validate resume parsing, section identification, and skill extraction logic. The dataset is obtained from Kaggle [4]. Only IT-related resumes are utilized for system evaluation to ensure relevance to technical job roles.

4.2 Resume Analysis and Feature Extraction

Uploaded resumes are converted from PDF to plain text and preprocessed using Natural Language Processing (NLP) techniques such as tokenization, stopword removal, and lemmatization [1, 3]. Section-based parsing and keyword matching are applied to extract relevant technical skills, tools, and experience-related information from the resume content.

4.3 Similarity Computation

Extracted resume content is compared with company-defined job requirements using TF-IDF vectorization and cosine similarity [2]. This process produces an interpretable similarity score representing the degree of alignment between a candidate profile and job requirements. The similarity score assists recruiters in structured candidate comparison and ranking.

4.4 Skill-Gap Feedback Mechanism

Missing skills are identified by comparing extracted resume skills with job requirements. These gaps are communicated to job seekers along with improvement suggestions. An external AI API (e.g., Gemini API) will be integrated to generate relevant course or learning recommendations based on identified skill gaps, providing personalized guidance for skill enhancement.

4.5 Non-Technical Skill Assessment

A quiz-based assessment module will be implemented to evaluate non-technical skills such as logical reasoning, communication, and problem-solving ability. Questions will be generated or selected through a structured question bank, and candidate responses will be automatically evaluated to produce a soft-skill score.

This soft-skill score may be combined with the similarity score to support overall candidate ranking and holistic evaluation.

Resume Analysis, Job Matching and Skill Assessment Workflow

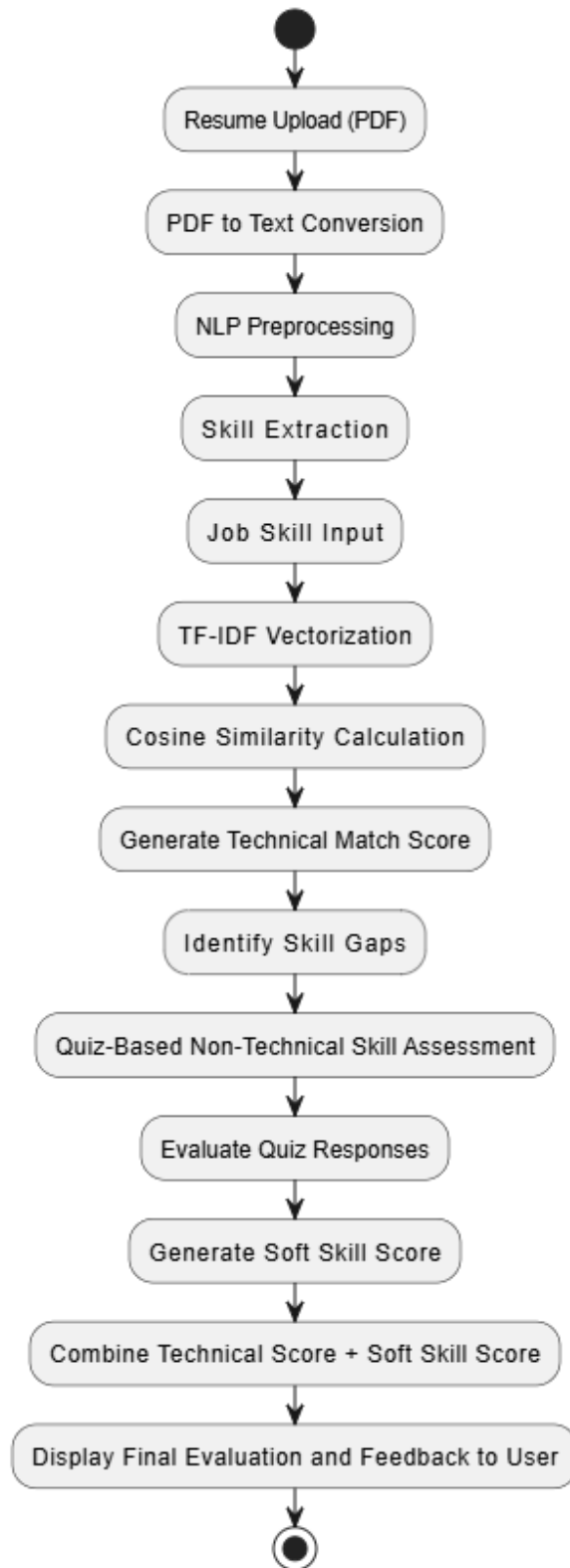


Figure 1: Resume Analysis, Job Matching, and Skill Assessment Workflow

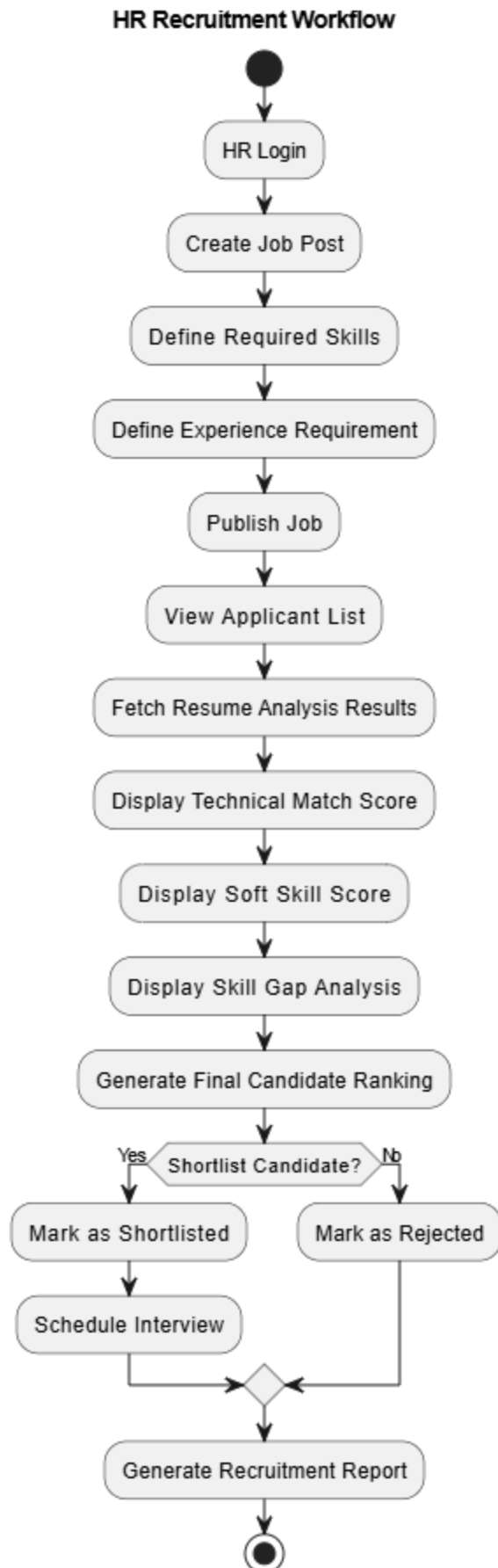


Figure 2: HR Recruitment Workflow
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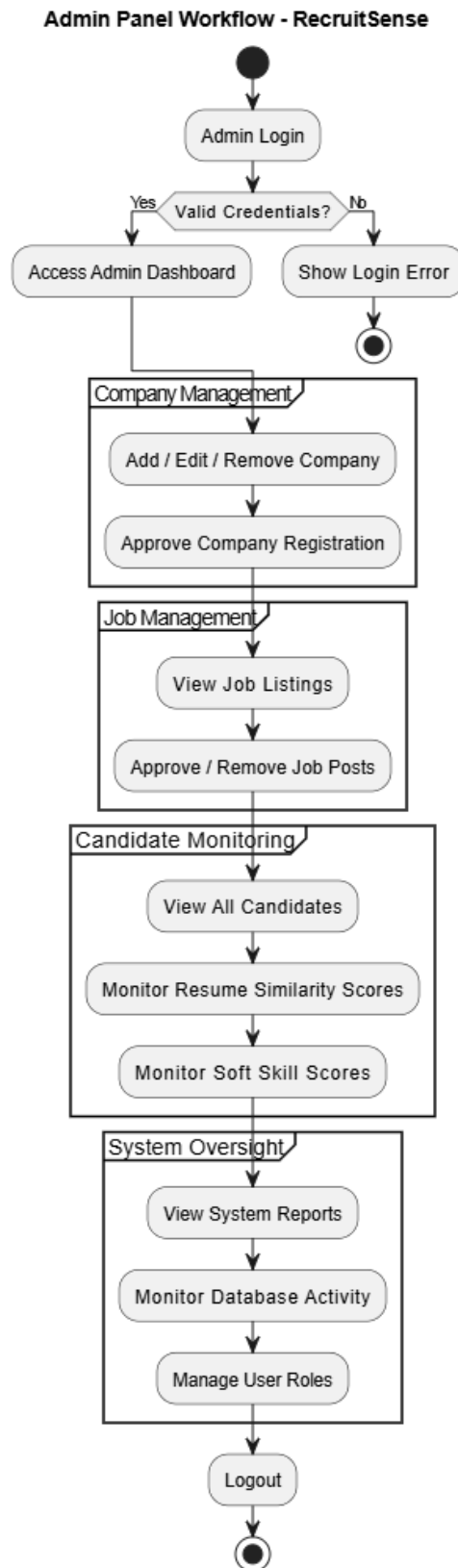


Figure 3: Admin Panel Workflow of the Proposed Recruitment System

5 Project Scope

5.1 In-Scope Features

- Web-based platform with Admin, Company, and Job Seeker panels
- Resume upload and parsing from PDF files
- Skill extraction and similarity calculation
- Candidate ranking and shortlist suggestions
- Skill-gap identification with learning guidance
- Quiz-based non-technical skill assessment (IQ, logical reasoning, communication)

5.2 Out-of-Scope Features

- Social media profile analysis
- Real-time integration with third-party learning platforms
- Video-based interview analysis or facial expression assessment
- Flutter mobile application development

6 Feasibility Study

6.1 Risks Involved

- Inaccurate skill extraction due to diverse and inconsistent resume formats
- Limitations of keyword and similarity-based techniques in fully capturing contextual meaning [2]

6.2 Resource Requirements

- Standard development environment (no GPU required)
- Local or cloud-based database for storing resumes and profiles

7 Solution Application Areas

- **Corporate HR Departments:** Assisting recruiters in managing high-volume IT job postings.
- **University Placement Offices:** Helping fresh graduates understand their market readiness.
- **Recruitment Agencies:** Streamlining the initial shortlisting process for various clients.

8 Comparative Analysis

This section compares existing recruitment platforms with the proposed system to highlight key functional differences.

Table 1: Comparative Analysis of Existing Platforms and Proposed System

Aspect	LinkedIn	Glassdoor	Indeed	Traditional ATS	Proposed System
Primary Purpose	Professional networking & visibility	Employer branding, reviews, salaries	Job search & application aggregation	Application tracking & workflow management	Recruitment decision-support system
Resume Evaluation	Mostly manual	Not supported	Basic screening questions	Keyword-based filtering	NLP-based structured resume analysis
Skill Extraction	Limited / implicit	Not available	Limited (employer-defined)	Basic keyword matching	Automated skill extraction using NLP
Job-Resume Matching	Basic relevance	Not supported	Basic relevance ranking	Rule/keyword-based matching	TF-IDF vectorization + cosine similarity
Non-Technical Skill Assessment	Not provided	Not provided	Not provided	Rare / external tools	Integrated quiz-based soft skill evaluation
Skill Gap Feedback	Not provided	Not provided	Not provided	Not provided	Explicit skill-gap identification
Explainable Scoring	Not available	Not available	Not available	Limited	Transparent similarity and combined evaluation scores
Candidate Ranking	Manual recruiter-driven	Not applicable	Basic ranking	Rule-based ranking	Similarity + soft-skill based ranking support
Support for Students	Generic	Limited	Generic	Limited	Skill-focused guidance and structured feedback
Decision Transparency	Low	Low	Low	Medium	High (explainable evaluation logic)
Hiring Automation	None	None	Partial (screening filters)	Partial	Decision-support only (human-in-the-loop)

9 Tools / Technology

Table 2: Tools and Technologies Used in the Proposed System

Category	Tool / Technology	Purpose / Usage
Frontend Development	ReactJS	To develop responsive and interactive user interfaces for Admin, Company, and Job Seeker panels
Backend Development	Laravel (PHP)	To handle business logic, authentication, authorization, API routing, and database operations
AI & NLP Processing	Python	To implement resume parsing, text preprocessing, skill extraction, and similarity computation
NLP Library	SpaCy	To perform tokenization, lemmatization, and basic Named Entity Recognition (NER)
Text Vectorization	Scikit-learn (TF-IDF)	To convert resume and job descriptions into numerical vectors for similarity analysis
Similarity Measurement	Cosine Similarity	To compute matching score between resume and job description
AI Service Framework	Flask	To deploy NLP and similarity logic as a lightweight API service connected to Laravel backend
PDF Processing	pdfplumber / PyMuPDF	To extract raw text from uploaded PDF resumes
Database	MySQL	To store user profiles, resumes, job descriptions, similarity scores, and skill-gap results
Dataset Source	Kaggle Resume Dataset	To validate and test resume parsing and skill extraction across different resume formats (not for deep model training)
Development Environment	VS Code	For code development and debugging
Version Control	Git / GitHub	To manage source code and track changes during development

10 Team Expertise

The project team comprises members with complementary technical expertise in frontend development, backend systems, and AI-based data processing. The division of responsibilities ensures balanced system architecture, efficient module development, and effective integration of the AI components within the web platform.

- **Haseeb Ahmed – Frontend & System Integration**

- Strong experience in web application development using HTML, CSS, JavaScript, and ReactJS [4].
- Skilled in designing responsive and user-friendly interfaces for multi-panel systems including Admin, Company, and Job Seeker dashboards.
- Experience in frontend–backend integration through API communication and handling dynamic data rendering.
- Familiar with implementing role-based access control at the interface level.
- Contributed to UI wireframing, workflow visualization, and overall system usability enhancement.

- **Hamza Zahoor – Backend & AI Support**

- Experience in backend development using Laravel and MySQL, including authentication, authorization, and database schema design.
- Developed RESTful APIs to integrate AI modules with the main web application [9].
- Proficient in Python programming for implementing Natural Language Processing tasks such as resume parsing, text preprocessing, and skill extraction [1, 3].
- Implemented TF-IDF vectorization and cosine similarity for job-resume matching [2].
- Responsible for integrating the AI evaluation module and ensuring proper system-level data flow between components.

The collaborative efforts of both team members ensure that the system is not only technically functional but also structured, scalable, and aligned with the objectives of AI-assisted recruitment decision support.

11 Milestones

The project development is divided into structured phases to ensure systematic implementation, proper documentation, and timely completion of all modules.

Table 3: Project Development Milestones

Phase No.	Time Period	Major Tasks / Activities
01	Feb – March 2026	Requirement analysis, literature review, proposal refinement, scope finalization, and preparation of initial documentation including Introduction, Objectives, and Problem Description.
02	March 2026	System architecture design, database schema creation, UI wireframes, workflow diagrams, and design documentation updates.
03	Apr – May 2026	Frontend development of Admin, Company, and Job Seeker panels, along with UI testing and documentation.
04	Jun – July 2026	Backend development including Laravel APIs, authentication mechanisms, role-based access control, database integration, and backend documentation.
05	Aug – Sep 2026	AI & NLP module development including resume parsing, skill extraction, job matching using TF-IDF and cosine similarity, and AI methodology documentation.
06	Sep 2026	System integration, module testing, AI evaluation, performance analysis, and preparation of testing and results documentation.
07	Oct 2026	Final review, report consolidation, formatting adjustments, proofreading, final submission, and presentation preparation.

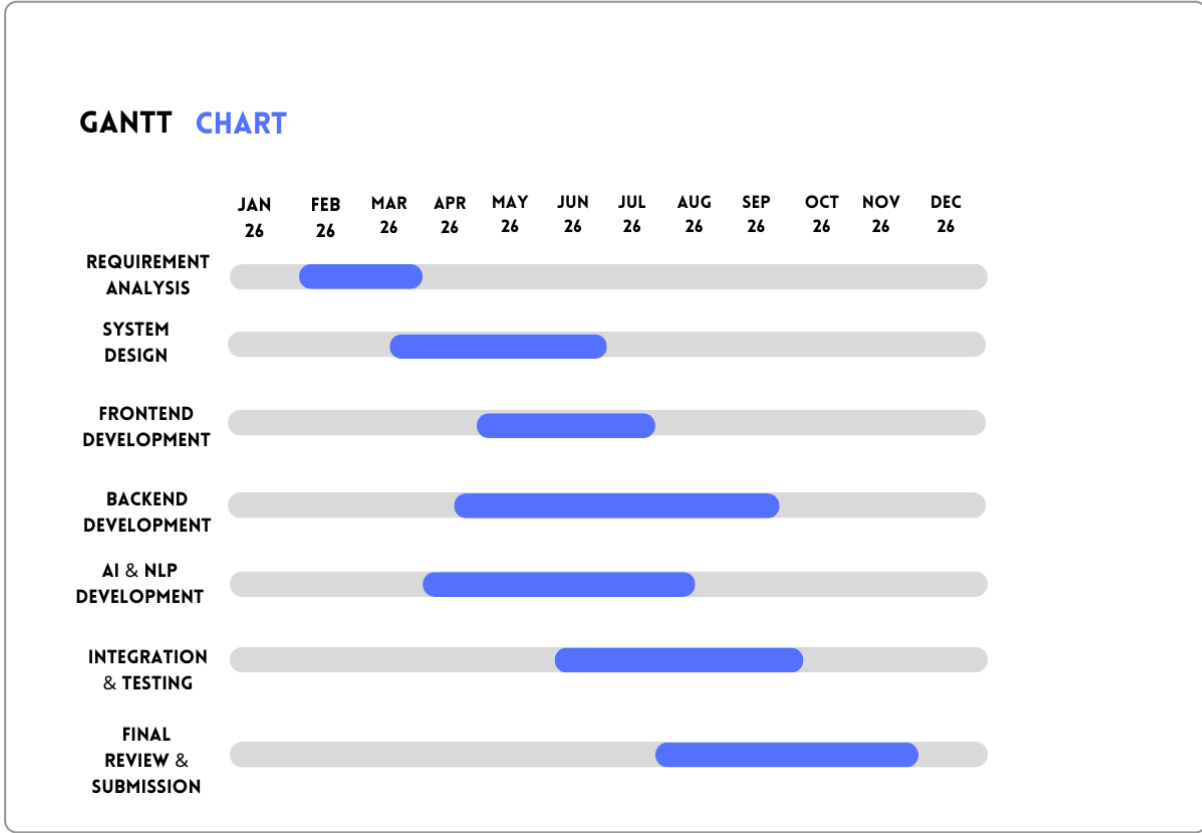


Figure 4: Project Development Gantt Chart

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